Chapter 8. Conformal Mappings

Conformal equivalence, Schwarz lemma and automorphisms, Riemann mapping theorem, Schwarz–Christoffel

Contents

1 Conformal equivalence and examples **2**

1.1 The disc and upper half-plane . . . 3

1.2 Further examples . . . 3

1.3 The Dirichlet problem in a strip . . . 4

2 The Schwarz lemma; automorphisms of the disc and upper half-plane **5**

2.1 Automorphisms of the disc . . . 6

2.2 Automorphisms of the upper half-plane . . . 7

3 The Riemann mapping theorem **7**

3.1 Necessary conditions and statement of the theorem . . . 7

3.2 Montel's theorem . . . 8

3.3 Proof of the Riemann mapping theorem . . . 9

4 Conformal mappings onto polygons **10**

4.1 Some examples . . . 10

4.2 The Schwarz–Christoffel integral . . . 11

4.3 Boundary behavior . . . 11

4.4 The mapping formula . . . 12

4.5 Return to elliptic integrals . . . 14

Overview

This chapter is geometric and "global": we study mapping properties of holomorphic functions, motivated by the question

> *Given two open sets U and V in $\mathbb{C}$, does there exist a holomorphic bijection between them?*

(The inverse is automatically holomorphic.) The key case is $V = \mathbb{D}$: *when is there a holomorphic bijection $\Omega \to \mathbb{D}$?* (For $V = \mathbb{C}$ only $U = \mathbb{C}$ works.) We give explicit maps and apply them to the Dirichlet problem, determine the automorphisms of $\mathbb{D}$ and $\mathbb{H}$ by the Schwarz lemma, and prove the *Riemann mapping theorem*: every proper simply connected Ω is conformally equivalent to $\mathbb{D}$, with no assumption on $\partial\Omega$. For polygons the map is the *Schwarz–Christoffel formula*; for rectangles it involves elliptic integrals, leading to the doubly periodic functions of Chapter 9.

1 Conformal equivalence and examples

Motivation. Regions are "the same" when a holomorphic bijection maps one onto the other; such maps transport problems to the disc or half-plane.

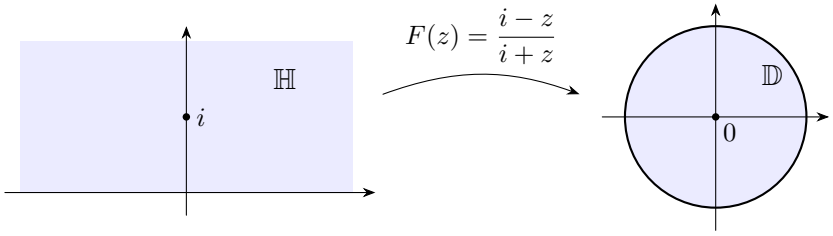


Figure. *The Cayley-type map $F\colon \mathbb{H} \to \mathbb{D}$, real axis onto the circle minus -1, $i \mapsto 0$.*

A bijective holomorphic $f : U \to V$ is a **conformal map** or **biholomorphism**; then U and V are **conformally equivalent** (**biholomorphic**).

Proposition 1.1

If $f : U \to V$ is holomorphic and injective, then $f'(z) \neq 0$ on U. So the inverse of f on its range, in particular of a conformal map, is holomorphic.

Proof. Suppose $f'(z_0) = 0$. Near z_0, $f(z) - f(z_0) = a(z - z_0)^k + G(z)$, $a \neq 0$, $k \geq 2$, G vanishing to order $k + 1$. For small w, $f(z) - f(z_0) - w = F(z) + G(z)$ with $F(z) = a(z - z_0)^k - w$; on a small circle $|G| < |F|$, and F has at least two zeros inside, so Rouché gives $f - f(z_0) - w$ at least two zeros, distinct as $f' \neq 0$ near z_0 ($z \neq z_0$). So f is not injective.

Let $g = f^{-1}$ on the range V, $w_0 = f(z_0)$, $w = f(z)$ close to w_0, $w \neq w_0$. Then

$$\frac{g(w) - g(w_0)}{w - w_0} = \frac{1}{\frac{w-w_0}{g(w)-g(w_0)}} = \frac{1}{\frac{f(z)-f(z_0)}{z-z_0}}.$$

Since $f'(z_0) \neq 0$, letting $z \to z_0$ shows g is holomorphic at w_0 with $g'(w_0) = 1/f'(g(w_0))$. ∎

So U, V are conformally equivalent iff there are holomorphic $f : U \to V$, $g : V \to U$ inverse to each other. (Some authors call f conformal when merely $f' \neq 0$, e.g. z^2 on $\mathbb{C} - \{0\}$; such f is a local bijection preserving angles.)

1.1 The disc and upper half-plane

The **upper half-plane** is $\mathbb{H} = \{z \in \mathbb{C} : \mathrm{Im}(z) > 0\}$. Let

$$F(z) = \frac{i - z}{i + z} \qquad \text{and} \qquad G(w) = i\frac{1 - w}{1 + w}.$$

Theorem 1.2

The map $F : \mathbb{H} \to \mathbb{D}$ is a conformal map with inverse $G : \mathbb{D} \to \mathbb{H}$.

Proof. Both maps are holomorphic in their domains. Points of $\mathbb{H}$ are closer to i than to $-i$, so $|F(z)| < 1$. For $w = u + iv \in \mathbb{D}$,

$$\mathrm{Im}(G(w)) = \mathrm{Re}\left(\frac{1 - u - iv}{1 + u + iv}\right) = \mathrm{Re}\,\frac{(1 - u - iv)(1 + u - iv)}{(1 + u)^2 + v^2} = \frac{1 - u^2 - v^2}{(1 + u)^2 + v^2} > 0.$$

Finally

$$F(G(w)) = \frac{i - i\frac{1-w}{1+w}}{i + i\frac{1-w}{1+w}} = \frac{1 + w - 1 + w}{1 + w + 1 - w} = w,$$

and similarly $G(F(z)) = z$. ∎

F is holomorphic on $\mathbb{C}$ except at $-i$, hence continuous on $\mathbb{R} = \partial\mathbb{H}$, and $|F(x)| = 1$ for real x. With $x = \tan t$, $t \in (-\pi/2, \pi/2)$,

$$F(x) = \frac{1 - x^2}{1 + x^2} + i\frac{2x}{1 + x^2} = \cos 2t + i \sin 2t = e^{2it},$$

so F maps $\mathbb{R}$ onto the circle minus -1, traversed from -1 first through the lower half; -1 corresponds to the point at infinity of $\mathbb{H}$.

Remark

Maps $z \mapsto \frac{az+b}{cz+d}$ (denominator not a multiple of the numerator) are **fractional linear transformations**; they give the automorphisms of $\mathbb{D}$ and $\mathbb{H}$ (Theorems 2.2, 2.4).

1.2 Further examples

Example 1

Translations and dilations. $z \mapsto z + h$ is conformal on $\mathbb{C}$ (and on $\mathbb{H}$ for real h); $z \mapsto cz$, $c \neq 0$, is conformal on $\mathbb{C}$: a rotation for $c = e^{i\varphi}$, a dilation for $c > 0$, a dilation by $|c|$ followed by rotation by π for $c < 0$.

Example 2

$z \mapsto z^n$ maps $\{0 < \arg z < \pi/n\}$ conformally onto $\mathbb{H}$ (inverse $w^{1/n}$, principal branch). For $0 < \alpha < 2$, $f(z) = z^\alpha$ maps $\mathbb{H}$ onto $S = \{0 < \arg w < \alpha\pi\}$: deleting the positive real axis, $f(re^{i\theta}) = r^\alpha e^{i\alpha\theta}$, $0 < \theta < \pi$, with inverse $w^{1/\alpha}$. With Example 1, $\mathbb{H}$ maps onto any infinite sector. On the boundary, $(-\infty, 0]$ goes onto the ray $\arg z = \alpha\pi$ (from $\infty e^{i\alpha\pi}$ to 0) and $[0, \infty)$ onto $[0, \infty)$.

Example 3

$f(z) = \frac{1+z}{1-z}$ maps the upper half-disc $\{|z| < 1,\ y > 0\}$ conformally onto the first quadrant $\{u > 0, v > 0\}$: for $z = x + iy$,

$$f(z) = \frac{1-(x^2+y^2)}{(1-x)^2+y^2} + i\frac{2y}{(1-x)^2+y^2}.$$

The inverse $g(w) = \frac{w-1}{w+1}$ maps the first quadrant into $\mathbb{D}$ (w is nearer 1 than -1) with $\operatorname{Im} g > 0$. On the boundary, $f(e^{i\theta}) = \frac{e^{-i\theta/2}+e^{i\theta/2}}{e^{-i\theta/2}-e^{i\theta/2}} = \frac{i}{\tan(\theta/2)}$ runs down the imaginary axis from ∞ to 0 as θ goes from 0 to π; f maps $(-1,1)$ increasingly onto $(0,\infty)$, $f(0) = 1$.

Example 4

$\log z$ (branch deleting the negative imaginary axis, $\log(re^{i\theta}) = \log r + i\theta$, $-\pi/2 < \theta < 3\pi/2$) maps $\mathbb{H}$ onto $\{0 < v < \pi\}$, inverse e^w; $(-\infty, 0)$ goes to the line $v = \pi$ (from $\infty + i\pi$ to $-\infty + i\pi$), $(0,\infty)$ onto $\mathbb{R}$.

Example 5

$\log z$ also maps the half-disc $\{|z| < 1, y > 0\}$ onto $\{u < 0,\ 0 < v < \pi\}$: $(0,1) \to (-\infty, 0)$, the upper half-circle to $[0, i\pi]$, $(-1,0)$ to the half-line from $i\pi$ to $-\infty + i\pi$.

Example 6

$f(z) = e^{iz} = e^{-y}e^{ix}$ maps $\{-\pi/2 < x < \pi/2,\ y > 0\}$ onto $\{|w| < 1,\ u > 0\}$. Boundary: the right side goes from 0 to i, $[-\pi/2, \pi/2]$ along the half-circle from i to $-i$, the left side from $-i$ to 0. (Cf. the inverse of Example 5.)

Example 7

$f(z) = -\frac{1}{2}(z + 1/z)$ maps $\{|z| < 1, y > 0\}$ conformally onto $\mathbb{H}$. Boundary: $(0,1) \to (-\infty, -1)$; on $z = e^{i\theta}$, $f = -\cos\theta$ runs from -1 to 1; $(-1,0) \to (1,\infty)$.

Example 8

$f(z) = \sin z$ maps the half-strip $\{-\pi/2 < x < \pi/2,\ y > 0\}$ conformally onto $\mathbb{H}$. With $\zeta = e^{iz}$,

$$\sin z = \frac{e^{iz} - e^{-iz}}{2i} = -\frac{1}{2}\Big(i\zeta + \frac{1}{i\zeta}\Big),$$

so f is Example 6, then multiplication by i, then Example 7. The left side goes to $(-\infty, -1]$, $[-\pi/2, \pi/2]$ onto $[-1, 1]$, the right side to $[1,\infty)$.

1.3 The Dirichlet problem in a strip

The **Dirichlet problem** in Ω is

$$\triangle u = 0 \text{ in } \Omega, \qquad u = f \text{ on } \partial\Omega, \tag{1}$$

$\triangle = \partial^2/\partial x^2 + \partial^2/\partial y^2$, f given on $\partial\Omega$. Poisson kernels solve it for the disc and $\mathbb{H}$; conformal maps give the strip.

Lemma 1.3

Let V and U be open sets in $\mathbb{C}$ and $F : V \to U$ a holomorphic function. If $u : U \to \mathbb{C}$ is a harmonic function, then $u \circ F$ is harmonic on V.

Proof. The question is local; on a disc U, $u = \operatorname{Re} G$ with G holomorphic, so $u \circ F = \operatorname{Re}(G \circ F)$ is harmonic. ∎

Let $\Omega = \{x + iy : x \in \mathbb{R},\ 0 < y < 1\}$, with boundary data $u(x, 0) = f_0(x)$, $u(x, 1) = f_1(x)$, f_0, f_1 continuous and vanishing at infinity. Transfer the problem to $\mathbb{D}$ by

$$F(w) = \frac{1}{\pi} \log\Big(i\frac{1-w}{1+w}\Big) : \mathbb{D} \to \Omega, \qquad G(z) = \frac{i - e^{\pi z}}{i + e^{\pi z}} : \Omega \to \mathbb{D},$$

compositions of maps above, mutually inverse. F maps the lower half-circle onto $i + \mathbb{R}$ (from $i + \infty$ to $i - \infty$) and the upper one onto $\mathbb{R}$ (from $-\infty$ to ∞). Let $\tilde{f}_1(\varphi) = f_1(F(e^{i\varphi}) - i)$, $-\pi < \varphi < 0$, and $\tilde{f}_0(\varphi) = f_0(F(e^{i\varphi}))$, $0 < \varphi < \pi$; as f_0, f_1 vanish at infinity, $\tilde{f}$ (0 at $\varphi = \pm\pi, 0$) is continuous on the circle. The disc solution is the Poisson integral, for $w = re^{i\theta}$,

$$\tilde{u}(w) = \frac{1}{2\pi}\int_{-\pi}^{\pi} P_r(\theta - \varphi)\tilde{f}(\varphi)\, d\varphi = \frac{1}{2\pi}\int_{-\pi}^{0} P_r(\theta - \varphi)\tilde{f}_1(\varphi)\, d\varphi + \frac{1}{2\pi}\int_{0}^{\pi} P_r(\theta - \varphi)\tilde{f}_0(\varphi)\, d\varphi,$$

$P_r(\gamma) = \frac{1-r^2}{1-2r\cos\gamma+r^2}$ the Poisson kernel. By Lemma 1.3, $u(z) = \tilde{u}(G(z))$ is harmonic in the strip, with the correct boundary values. For $z = iy$, $0 < y < 1$, with $re^{i\theta} = G(iy)$, a change of variables gives

$$\frac{1}{2\pi}\int_{0}^{\pi} P_r(\theta-\varphi)\tilde{f}_0(\varphi)\, d\varphi = \frac{\sin \pi y}{2}\int_{-\infty}^{\infty} \frac{f_0(t)}{\cosh \pi t - \cos \pi y}\, dt, \qquad \frac{1}{2\pi}\int_{-\pi}^{0} P_r(\theta-\varphi)\tilde{f}_1(\varphi)\, d\varphi = \frac{\sin \pi y}{2}\int_{-\infty}^{\infty} \frac{f_1(t}{\cosh \pi t +}$$

A solution vanishing at infinity is unique, so translating the boundary data by x translates the solution, and

$$u(x, y) = \frac{\sin \pi y}{2}\Big(\int_{-\infty}^{\infty} \frac{f_0(x-t)}{\cosh \pi t - \cos \pi y}\, dt + \int_{-\infty}^{\infty} \frac{f_1(x-t)}{\cosh \pi t + \cos \pi y}\, dt\Big),$$

convolutions with f_0 and f_1. At $y = 1/2$ the kernel is $1/\cosh \pi t$, its own Fourier transform (Example 3 in Section 2 of Chapter 3).

Remarks about the Dirichlet problem. If $F : \mathbb{D} \to \Omega$ is conformal and extends to a continuous bijection $\partial\mathbb{D} \to \partial\Omega$, then with $\tilde{f} = f \circ F$ on the circle and $\tilde{u}(re^{i\theta}) = \frac{1}{2\pi}\int_0^{2\pi} P_r(\theta - \varphi)\tilde{f}(e^{i\varphi})\, d\varphi$, $u = \tilde{u} \circ F^{-1}$ should solve (1). This requires:

(i) existence of a conformal map $\Phi = F^{-1} : \Omega \to \mathbb{D}$, settled in general by the Riemann mapping theorem;

(ii) its extension to a continuous bijection of the boundaries, needing some regularity of $\partial\Omega$ (polygons: Section 4). Riemann reversed this: writing $\Phi(z) = (z - z_0)e^{H(z)}$, $|\Phi| = 1$ on $\partial\Omega$ means $u = \operatorname{Re} H$ solves the Dirichlet problem with data $\log(1/|z - z_0|)$ (the Green's function). This needs bijectivity of Φ, boundary regularity, and a solution of the Dirichlet problem (Riemann's "Dirichlet principle" is delicate). Section 3 proves existence differently.

2 The Schwarz lemma; automorphisms of the disc and upper half-plane

Motivation. The Schwarz lemma forces the automorphisms of $\mathbb{D}$ and $\mathbb{H}$ to be Möbius transformations.

A **rotation** is a map $z \mapsto cz$ with $|c| = 1$, $c = e^{i\varphi}$, φ the angle of rotation (defined modulo 2π).

Lemma 2.1 (Schwarz lemma)

Let $f : \mathbb{D} \to \mathbb{D}$ be holomorphic with $f(0) = 0$. Then (i) $|f(z)| \le |z|$ for all $z \in \mathbb{D}$.
(ii) If for some $z_0 \neq 0$ we have $|f(z_0)| = |z_0|$, then f is a rotation.
(iii) $|f'(0)| \le 1$, and if equality holds, then f is a rotation.

Proof. $f(0) = 0$, so $f(z)/z$ is holomorphic in $\mathbb{D}$. On $|z| = r < 1$, $|f(z)/z| \le 1/r$, hence on $|z| \le r$ by the maximum modulus principle; $r \to 1$ gives (i). For (ii), $f(z)/z$ attains its maximum inside $\mathbb{D}$, so it is a constant c, $|c| = 1$. For (iii), $g = f(z)/z$ has $|g| \le 1$ and $g(0) = f'(0)$; if $|f'(0)| = 1$, g is constant by the maximum principle, $f(z) = cz$, $|c| = 1$. ■

2.1 Automorphisms of the disc

A conformal map of Ω onto itself is an **automorphism**; they form the group $\mathrm{Aut}(\Omega)$. Rotations $r_\theta : z \mapsto e^{i\theta} z$ are automorphisms of $\mathbb{D}$. More interesting are

$$\psi_\alpha(z) = \frac{\alpha - z}{1 - \overline{\alpha} z}, \qquad \alpha \in \mathbb{C},\ |\alpha| < 1.$$

ψ_α is holomorphic in $\mathbb{D}$. If $z = e^{i\theta}$, $\psi_\alpha(e^{i\theta}) = \frac{\alpha - e^{i\theta}}{e^{i\theta}(e^{-i\theta} - \overline{\alpha})} = -e^{-i\theta}\frac{w}{\overline{w}}$ with $w = \alpha - e^{i\theta}$, so $|\psi_\alpha| = 1$ on the circle and $|\psi_\alpha| < 1$ in $\mathbb{D}$ by the maximum modulus principle. Moreover

$$(\psi_\alpha \circ \psi_\alpha)(z) = \frac{\alpha - \frac{\alpha - z}{1 - \overline{\alpha} z}}{1 - \overline{\alpha}\frac{\alpha - z}{1 - \overline{\alpha} z}} = \frac{\alpha - |\alpha|^2 z - \alpha + z}{1 - \overline{\alpha} z - |\alpha|^2 + \overline{\alpha} z} = \frac{(1 - |\alpha|^2) z}{1 - |\alpha|^2} = z,$$

so ψ_α is its own inverse; and it interchanges 0 and α: $\psi_\alpha(0) = \alpha$, $\psi_\alpha(\alpha) = 0$.

Theorem 2.2

If f is an automorphism of the disc, then there exist $\theta \in \mathbb{R}$ and $\alpha \in \mathbb{D}$ such that

$$f(z) = e^{i\theta} \frac{\alpha - z}{1 - \overline{\alpha} z}.$$

Proof. There is a unique $\alpha \in \mathbb{D}$ with $f(\alpha) = 0$. Let $g = f \circ \psi_\alpha$; then $g(0) = 0$ and by the Schwarz lemma

$$|g(z)| \le |z| \qquad \text{for all } z \in \mathbb{D}. \tag{2}$$

Also $g^{-1}(0) = 0$, so $|g^{-1}(w)| \le |w|$; with $w = g(z)$,

$$|z| \le |g(z)| \qquad \text{for all } z \in \mathbb{D}. \tag{3}$$

By (2), (3), $|g(z)| = |z|$, so $g(z) = e^{i\theta} z$ by the Schwarz lemma. Replacing z by $\psi_\alpha(z)$ and using $\psi_\alpha \circ \psi_\alpha = \mathrm{id}$, $f(z) = e^{i\theta}\psi_\alpha(z)$. ■

Corollary 2.3

The only automorphisms of the unit disc that fix the origin are the rotations.

$\mathrm{Aut}(\mathbb{D})$ acts *transitively* ($\psi_\beta \circ \psi_\alpha$ maps α to β) and is "almost" isomorphic to $SU(1,1)$, the 2×2 complex matrices preserving $\langle Z, W\rangle = z_1\overline{w_1} - z_2\overline{w_2}$.

2.2 Automorphisms of the upper half-plane

With $F : \mathbb{H} \to \mathbb{D}$ from Section 1, define $\Gamma : \mathrm{Aut}(\mathbb{D}) \to \mathrm{Aut}(\mathbb{H})$ by "conjugation by F": $\Gamma(\varphi) = F^{-1} \circ \varphi \circ F$. Γ is a bijection with inverse $\Gamma^{-1}(\psi) = F \circ \psi \circ F^{-1}$, and it preserves composition:

$$\Gamma(\varphi_1 \circ \varphi_2) = F^{-1} \circ \varphi_1 \circ F \circ F^{-1} \circ \varphi_2 \circ F = \Gamma(\varphi_1) \circ \Gamma(\varphi_2).$$

So $\mathrm{Aut}(\mathbb{D}) \cong \mathrm{Aut}(\mathbb{H})$. Let $SL_2(\mathbb{R}) = \left\{ M = \begin{pmatrix} a & b \\ c & d \end{pmatrix} : a, b, c, d \in \mathbb{R},\ \det(M) = ad - bc = 1 \right\}$, the **special linear group**, and for $M \in SL_2(\mathbb{R})$ let $f_M(z) = \frac{az+b}{cz+d}$.

Theorem 2.4

Every automorphism of $\mathbb{H}$ takes the form f_M for some $M \in SL_2(\mathbb{R})$. Conversely, every map of this form is an automorphism of $\mathbb{H}$.

Proof. Write $G = SL_2(\mathbb{R})$.

Step 1. If $M \in G$, f_M maps $\mathbb{H}$ to itself:

$$\mathrm{Im}(f_M(z)) = \frac{(ad - bc)\,\mathrm{Im}(z)}{|cz + d|^2} = \frac{\mathrm{Im}(z)}{|cz + d|^2} > 0 \qquad (z \in \mathbb{H}). \tag{4}$$

Step 2. $f_M \circ f_{M'} = f_{MM'}$ (a straightforward calculation). Hence each f_M is an automorphism, with holomorphic inverse $f_{M^{-1}}$: $(f_M \circ f_{M^{-1}})(z) = f_I(z) = z$.

Step 3. G acts transitively on $\mathbb{H}$: it suffices to map $z \in \mathbb{H}$ to i. With $d = 0$ in (4), $\mathrm{Im}(f_M(z)) = \mathrm{Im}(z)/|cz|^2$; choose $c \in \mathbb{R}$ making this 1, $M_1 = \begin{pmatrix} 0 & -c^{-1} \\ c & 0 \end{pmatrix}$, then translate by $M_2 = \begin{pmatrix} 1 & b \\ 0 & 1 \end{pmatrix}$, $b \in \mathbb{R}$; $M = M_2 M_1$.

Step 4. For $\theta \in \mathbb{R}$, $M_\theta = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix} \in G$, and $F \circ f_{M_\theta} \circ F^{-1}$ is the rotation of angle -2θ in the disc, since $F \circ f_{M_\theta} = e^{-2i\theta} F(z)$ (easily verified).

Step 5. Let $f \in \mathrm{Aut}(\mathbb{H})$ with $f(\beta) = i$, and $N \in G$ with $f_N(i) = \beta$. Then $g = f \circ f_N$ fixes i, so $F \circ g \circ F^{-1}$ is an automorphism of $\mathbb{D}$ fixing 0, hence a rotation; by Step 4, $F \circ g \circ F^{-1} = F \circ f_{M_\theta} \circ F^{-1}$ for some θ. So $g = f_{M_\theta}$ and $f = f_{M_\theta N^{-1}}$. ∎

M and $-M$ give the same f_M; identifying them gives the **projective special linear group** $PSL_2(\mathbb{R})$, which is isomorphic to $\mathrm{Aut}(\mathbb{H})$.

3 The Riemann mapping theorem

3.1 Necessary conditions and statement of the theorem

For $\Omega = \mathbb{C}$ there is no conformal map to $\mathbb{D}$ (Liouville); Ω must be connected and simply connected, like $\mathbb{D}$. These conditions suffice. A subset of $\mathbb{C}$ is **proper** if it is non-empty and not $\mathbb{C}$.

Theorem 3.1 (Riemann mapping theorem)

Suppose Ω is proper and simply connected. If $z_0 \in \Omega$, then there exists a unique conformal map $F : \Omega \to \mathbb{D}$ such that $F(z_0) = 0$ and $F'(z_0) > 0$.

Corollary 3.2

Any two proper simply connected open subsets in $\mathbb{C}$ are conformally equivalent.

The corollary follows via $\mathbb{D}$. *Uniqueness*: $H = F \circ G^{-1} \in \mathrm{Aut}(\mathbb{D})$ fixes 0, so $H(z) = e^{i\theta}z$, and $H'(0) > 0$ gives $F = G$. For existence, among injective holomorphic $f : \Omega \to \mathbb{D}$ with $f(z_0) = 0$ take one maximizing $|f'(z_0)|$; it fills $\mathbb{D}$. We need limits of sequences of functions.

3.2 Montel's theorem

A family $\mathcal{F}$ of holomorphic functions on Ω is **normal** if every sequence in $\mathcal{F}$ has a subsequence converging uniformly on compact subsets (limit not necessarily in $\mathcal{F}$); **uniformly bounded on compact subsets** if for each compact K some B bounds all $|f|$, $f \in \mathcal{F}$, on K; **equicontinuous** on compact K if for every $\epsilon > 0$ some $\delta > 0$ gives $|f(z) - f(w)| < \epsilon$ for all $f \in \mathcal{F}$ and $z, w \in K$, $|z-w| < \delta$. (Differentiable functions on $[0,1]$ with uniformly bounded derivatives are equicontinuous; x^n on $[0,1]$ are not.)

Theorem 3.3 (Montel)

Suppose $\mathcal{F}$ is a family of holomorphic functions on Ω that is uniformly bounded on compact subsets of Ω. Then:
(i) $\mathcal{F}$ is equicontinuous on every compact subset of Ω.
(ii) $\mathcal{F}$ is a normal family.

Part (i) uses the Cauchy integral formula (contrast $\sin(nx)$ on $(0,1)$: bounded, not equicontinuous); part (ii) is Arzelà–Ascoli. A sequence of compact $K_\ell \subset \Omega$ is an **exhaustion** if (a) K_ℓ lies in the interior of $K_{\ell+1}$, and (b) every compact $K \subset \Omega$ lies in some K_ℓ; so $\Omega = \bigcup_\ell K_\ell$.

Lemma 3.4

Any open set Ω in the complex plane has an exhaustion.

Proof. If Ω is bounded, let K_ℓ be the set of points of Ω at distance $\geq 1/\ell$ from $\partial\Omega$. If Ω is unbounded, require also $|z| \leq \ell$ for $z \in K_\ell$. ∎

Proof of Montel's theorem. (i) Let $K \subset \Omega$ be compact and $r > 0$ with $3r < d(K, \partial\Omega)$, so $D_{3r}(z) \subset \Omega$ for $z \in K$. Let $z, w \in K$ with $|z - w| < r$ and γ the boundary circle of $D_{2r}(w)$. By Cauchy's integral formula

$$f(z) - f(w) = \frac{1}{2\pi i}\int_\gamma f(\zeta)\Big[\frac{1}{\zeta - z} - \frac{1}{\zeta - w}\Big]d\zeta, \qquad \Big|\frac{1}{\zeta - z} - \frac{1}{\zeta - w}\Big| = \frac{|z-w|}{|\zeta - z||\zeta - w|} \leq \frac{|z-w|}{r^2},$$

since $\zeta \in \gamma$ and $|z - w| < r$. Hence $|f(z) - f(w)| \leq \frac{1}{2\pi} 2\pi(2r)\frac{B|z-w|}{r^2}$, B the uniform bound for $\mathcal{F}$ on the compact set of points at distance $\leq 2r$ from K. So $|f(z) - f(w)| < C|z - w|$ for all $f \in \mathcal{F}$: equicontinuity.

(ii) Let $\{f_n\} \subset \mathcal{F}$, $K \subset \Omega$ compact, and $\{w_j\}$ dense in Ω. By boundedness, successive subsequences converge at w_1, w_2, …; the diagonal $g_n = f_{n,n}$ converges at every w_j. Uniformly on K: given ϵ, take δ from equicontinuity, cover K by $D_\delta(w_1), \ldots, D_\delta(w_J)$, and choose N with $|g_m(w_j) - g_n(w_j)| < \epsilon$ for $n, m > N$, $j \leq J$. For $z \in K \cap D_\delta(w_j)$,

$$|g_n(z) - g_m(z)| \leq |g_n(z) - g_n(w_j)| + |g_n(w_j) - g_m(w_j)| + |g_m(w_j) - g_m(z)| < 3\epsilon.$$

Finally, with an exhaustion, diagonalize again: $\{g_{n,n}\}$ converges uniformly on every K_ℓ, hence on every compact subset. ∎

Proposition 3.5

If Ω is connected open and injective holomorphic f_n converge uniformly on compact subsets to f, then f is injective or constant.

Proof. Suppose $f(z_1) = f(z_2)$, $z_1 \neq z_2$. $g_n = f_n - f_n(z_1)$ vanishes only at z_1, and $g_n \to g = f - f(z_1)$. If $g \not\equiv 0$, z_2 is an isolated zero of g, so

$$1 \leq \frac{1}{2\pi i} \int_\gamma \frac{g'(\zeta)}{g(\zeta)}\, d\zeta,$$

for a small circle γ about z_2 with no other zeros of g on or inside it. As $1/g_n \to 1/g$, $g_n' \to g'$ uniformly on γ (Chapter 2, Theorem 5.3), $\frac{1}{2\pi i}\int_\gamma \frac{g_n'}{g_n} \to \frac{1}{2\pi i}\int_\gamma \frac{g'}{g}$; but the left side is 0 ($g_n \neq 0$ inside γ), a contradiction. So $g \equiv 0$ and f is constant. ■

3.3 Proof of the Riemann mapping theorem

Step 1. Ω is conformally equivalent to an open subset of $\mathbb{D}$ containing 0. Take $\alpha \notin \Omega$; on the simply connected Ω there is $f(z) = \log(z - \alpha)$, $e^f = z - \alpha$, so f is injective. Fix $w \in \Omega$: $f(z) \neq f(w) + 2\pi i$ (exponentiate), and $f(\Omega)$ stays away from $f(w) + 2\pi i$, since $f(z_n) \to f(w) + 2\pi i$ would give $z_n \to w$, so $f(z_n) \to f(w)$. Thus

$$F(z) = \frac{1}{f(z) - (f(w) + 2\pi i)}$$

is injective and bounded on Ω; translating and rescaling $F(\Omega)$ gives the claim.

Step 2. So assume $\Omega \subset \mathbb{D}$ is open with $0 \in \Omega$, and let

$$\mathcal{F} = \{f : \Omega \to \mathbb{D} \text{ holomorphic, injective, and } f(0) = 0\}.$$

$\mathcal{F} \neq \emptyset$ (identity) is uniformly bounded, and so are the $|f'(0)|$ (Cauchy inequalities). Let $s = \sup_{\mathcal{F}} |f'(0)| \geq 1$ and $|f_n'(0)| \to s$. By Montel a subsequence converges uniformly on compacta to f, non-constant as $s \geq 1$, hence injective (Proposition 3.5). $|f| \leq 1$ by continuity, $|f| < 1$ by maximum modulus, $f(0) = 0$: $f \in \mathcal{F}$ with $|f'(0)| = s$.

Step 3. f is surjective. Otherwise take $\alpha \in \mathbb{D} - f(\Omega)$ and $\psi_\alpha(z) = \frac{\alpha - z}{1 - \overline{\alpha} z}$. $U = (\psi_\alpha \circ f)(\Omega)$ is simply connected and omits 0, so $g(w) = e^{\frac{1}{2}\log w}$ is defined on U. Let

$$F = \psi_{g(\alpha)} \circ g \circ \psi_\alpha \circ f.$$

F is holomorphic, maps 0 to 0 ($g(\psi_\alpha(0)) = g(\alpha)$), maps Ω into $\mathbb{D}$, and is injective (each factor is). So $F \in \mathcal{F}$. With $h(w) = w^2$,

$$f = \psi_\alpha^{-1} \circ h \circ g^{-1} \circ \psi_{g(\alpha)}^{-1} \circ F = \Phi \circ F.$$

$\Phi = \psi_\alpha \circ h \circ \psi_{g(\alpha)}$ maps $\mathbb{D}$ into $\mathbb{D}$, $\Phi(0) = 0$, and is not injective (h is not), so $|\Phi'(0)| < 1$ by the Schwarz lemma. As $f'(0) = \Phi'(0)F'(0)$, $|f'(0)| < |F'(0)|$, contradicting maximality.

Finally multiply f by a constant of absolute value 1 so that $f'(0) > 0$. ■

Remark

Simple connectivity was used only for the logarithm and square root; *holomorphic simple connectivity* ($\int_\gamma f = 0$ for all holomorphic f and closed $\gamma \subset \Omega$) suffices. See Appendix B.

4 Conformal mappings onto polygons

For polygons the map is explicit: the *Schwarz–Christoffel formula*.

4.1 Some examples

Example 1

For $0 < \alpha < 2$, the map $f(z) = z^\alpha$ of $\mathbb{H}$ onto the sector $\{0 < \arg z < \alpha\pi\}$ (Section 1) can be written

$$z^\alpha = f(z) = \int_0^z f'(\zeta)\,d\zeta = \alpha \int_0^z \zeta^{-\beta}\,d\zeta, \qquad \alpha + \beta = 1,$$

integrated along any path in the closed upper half-plane. $\zeta^{-\beta}$ is integrable at 0 ($\beta < 1$), so $f(0) = 0$. For $x > 0$, $f'(x) = \alpha x^{\alpha-1} > 0$, not integrable at ∞, so $[0,\infty) \to [0,\infty)$. For $x < 0$, $f'(x) = -\alpha|x|^{\alpha-1}e^{i\pi\alpha}$, so $(-\infty, 0] \to (\infty e^{i\pi\alpha}, 0]$.

Example 2

For $z \in \mathbb{H}$ let $f(z) = \int_0^z \frac{d\zeta}{(1-\zeta^2)^{1/2}}$, the branch holomorphic in $\mathbb{H}$ and positive on $(-1,1)$; then $(1-\zeta^2)^{-1/2} = -i(\zeta^2-1)^{-1/2}$ for $\zeta > 1$ (similarly for $\zeta < -1$). $f(\pm 1) = \pm\pi/2$ and $f' > 0$ on $(-1,1)$, so $[-1,1] \to [-\pi/2, \pi/2]$; for $x > 1$, $f(x) = \frac{\pi}{2} + i\int_1^x \frac{dx}{(x^2-1)^{1/2}}$, which diverges, so $[1,\infty)$ goes onto the vertical half-line above $\pi/2$; similarly for $(-\infty, -1]$. In fact f inverts $\sin z$ (Example 8) and maps $\mathbb{H}$ conformally onto the half-strip.

Example 3

For fixed $0 < k < 1$, let

$$f(z) = \int_0^z \frac{d\zeta}{[(1-\zeta^2)(1-k^2\zeta^2)]^{1/2}}, \qquad z \in \mathbb{H},$$

the branch positive for $-1 < \zeta < 1$. Such integrals are **elliptic integrals** (variants arise in the arc length of an ellipse). With

$$K = \int_0^1 \frac{dx}{[(1-x^2)(1-k^2x^2)]^{1/2}}, \qquad K' = \int_1^{1/k} \frac{dx}{[(x^2-1)(1-k^2x^2)]^{1/2}},$$

f maps $\mathbb{R}$ onto the rectangle with vertices $-K$, K, $K+iK'$, $-K+iK'$. $f(\pm 1) = \pm K$, $f' > 0$ on $(-1,1)$: $[-1,1] \to [-K, K]$. For $1 < x < 1/k$, $f(x) = K + i\int_1^x \frac{d\xi}{[(\xi^2-1)(1-k^2\xi^2)]^{1/2}}$: $[1, 1/k] \to [K, K+iK']$; similarly $[-1/k, -1] \to [-K+iK', -K]$. For $x > 1/k$, $f(x) = K + iK' - \int_{1/k}^x \frac{dx}{[(x^2-1)(k^2x^2-1)]^{1/2}}$, and $\int_{1/k}^\infty \frac{dx}{[(x^2-1)(k^2x^2-1)]^{1/2}} = K$ ($x = 1/ku$), so $[1/k, \infty) \to [K+iK', iK']$; similarly $(-\infty, -1/k] \to [-K+iK', iK']$. ∞ goes to iK'.

Two problems: generalize (the Schwarz–Christoffel integral maps $\mathbb{R}$ to a polygonal line), and prove conformality from $\mathbb{H}$ onto the polygon's interior.

4.2 The Schwarz–Christoffel integral

Definition (Schwarz–Christoffel integral)

For real $A_1 < A_2 < \cdots < A_n$ and exponents $\beta_k < 1$ with $1 < \sum_{k=1}^n \beta_k$, let

$$S(z) = \int_0^z \frac{d\zeta}{(\zeta - A_1)^{\beta_1} \cdots (\zeta - A_n)^{\beta_n}}, \tag{5}$$

where $(z - A_k)^{\beta_k}$ is the branch in the plane slit along $\{A_k + iy : y \le 0\}$ that is positive for real $x > A_k$; thus $(x - A_k)^{\beta_k} = |x - A_k|^{\beta_k} e^{i\pi\beta_k}$ for real $x < A_k$.

The plane slit along the rays $\{A_k + iy : y \le 0\}$ is simply connected, so S is holomorphic there; the singularities are integrable ($\beta_k < 1$), so S is continuous up to $\mathbb{R}$, along any path avoiding the open slits. As $\prod_k |\zeta - A_k|^{-\beta_k} \le c|\zeta|^{-\sum \beta_k}$ with $\sum \beta_k > 1$, the integral converges at infinity and by Cauchy $a_\infty = \lim_{r\to\infty} S(re^{i\theta})$ exists independently of $0 \le \theta \le \pi$; let $a_k = S(A_k)$. (The cases $\sum \beta_k \le 1$ of Examples 1–2 need a modification, S then being unbounded.)

Proposition 4.1

Suppose $S(z)$ is given by (5).
(i) If $\sum_{k=1}^n \beta_k = 2$ and p is the polygon with vertices $a_1, \dots, a_n$ (in order), S maps $\mathbb{R}$ onto $p - \{a_\infty\}$, $a_\infty \in [a_n, a_1]$ being the image of ∞, and the interior angle at a_k is $\alpha_k\pi$, $\alpha_k = 1 - \beta_k$.
(ii) If $1 < \sum_{k=1}^n \beta_k < 2$, the image of the extended line is the $(n+1)$-gon $a_1, \dots, a_n, a_\infty$, with angle $\alpha_\infty\pi$ at a_∞, $\alpha_\infty = 1 - \beta_\infty$, $\beta_\infty = 2 - \sum_{k=1}^n \beta_k$.

Proof. Let $\sum \beta_k = 2$. For $A_k < x < A_{k+1}$,

$$S'(x) = \prod_{j\le k} (x - A_j)^{-\beta_j} \prod_{j>k} (x - A_j)^{-\beta_j}, \qquad \arg S'(x) = \arg \prod_{j>k} e^{-i\pi\beta_j} = -\pi \sum_{j>k} \beta_j,$$

constant on (A_k, A_{k+1}). As $S(x) = S(A_k) + \int_{A_k}^x S'(y)\,dy$, S moves along the segment $[a_k, a_{k+1}]$ ($[z, w] = \{(1-t)z + tw : t \in [0,1]\}$) at angle $-\pi \sum_{j>k} \beta_j$. For $x > A_n$, $S'(x) > 0$; for $x < A_1$, $\arg S'(x) = -2\pi$, so again $S' > 0$. So $[A_n, \infty)$ and $(-\infty, A_1]$ give $[a_n, a_\infty) \cup (a_\infty, a_1] = [a_n, a_1] - \{a_\infty\}$. The direction of $[a_k, a_{k+1}]$ exceeds that of $[a_{k-1}, a_k]$ by $\pi\beta_k$, so the interior angle at a_k is $\pi - \pi\beta_k = \alpha_k\pi$. The case $1 < \sum \beta_k < 2$ is similar. ∎

This does not settle the problem:

(1) for general n and A_k the image polygon need not be simple, nor S conformal on $\mathbb{H}$;

(2) it does not show that for a given polygonal region P, S (suitably modified) maps $\mathbb{H}$ conformally onto P. We now prove it does.

4.3 Boundary behavior

A **polygonal region** P is a bounded simply connected open set whose boundary is a closed polygonal line p (a *polygon*). For a conformal $f : U \to f(U)$,

$$\text{Area}(f(U)) = \iint_U |f'(z)|^2\,dx\,dy,$$

since the Jacobian of $w = f(z)$ is $|f'(z)|^2$ (Proposition 2.3 of Chapter 1).

Theorem 4.2

A conformal map $F : \mathbb{D} \to P$ extends to a continuous bijection $\overline{\mathbb{D}} \to \overline{P}$; in particular $\partial\mathbb{D}$ maps bijectively onto the polygon p.

Lemma 4.3

Let C_r $(0 < r < 1/2)$ be the circle of radius r about $z_0 \in \partial\mathbb{D}$, and $z_r, z_r' \in \mathbb{D} \cap C_r$ for all small r. With $\rho(r) = |F(z_r) - F(z_r')|$, there are $r_n \to 0$ with $\rho(r_n) \to 0$.

Proof. If not, there are $c > 0$, $0 < R < 1/2$ with $c \le \rho(r)$ for $0 < r \le R$. Now $F(z_r) - F(z_r') = \int_\gamma F'(\zeta)\,d\zeta$, γ the arc of C_r in $\mathbb{D}$ joining z_r, z_r'; parametrizing it by $z_0 + re^{i\theta}$, $\theta_1(r) \le \theta \le \theta_2(r)$,

$$\rho(r) \le \int_{\theta_1(r)}^{\theta_2(r)} |F'(z)| r\,d\theta \le \Big(\int_{\theta_1(r)}^{\theta_2(r)} |F'(z)|^2 r\,d\theta\Big)^{1/2} \Big(\int_{\theta_1(r)}^{\theta_2(r)} r\,d\theta\Big)^{1/2}$$

by Cauchy–Schwarz. Squaring and dividing by r, $\frac{\rho(r)^2}{r} \le 2\pi \int_{\theta_1(r)}^{\theta_2(r)} |F'(z)|^2 r\,d\theta$. Integrating from 0 to R,

$$c^2 \int_0^R \frac{dr}{r} \le 2\pi \int_0^R \int_{\theta_1(r)}^{\theta_2(r)} |F'(z)|^2 r\,d\theta\,dr \le 2\pi \iint_{\mathbb{D}} |F'(z)|^2\,dx\,dy = 2\pi\,\mathrm{Area}(P).$$

The left side is infinite and the right side finite—a contradiction. ∎

Lemma 4.4

Let z_0 be a point on the unit circle. Then $F(z)$ tends to a limit as z approaches z_0 within the unit disc.

Proof. Otherwise $z_n, z_n' \to z_0$ in $\mathbb{D}$ with $F(z_n) \to \zeta$, $F(z_n') \to \zeta'$, $\zeta \ne \zeta'$, both on p (conformality). Take disjoint discs D, D' about ζ, ζ' at distance $d > 0$, and continuous curves $\Lambda \subset D \cap P$, $\Lambda' \subset D' \cap P$ through $F(z_n)$, resp. $F(z_n')$, for all large n, ending at ζ, ζ'. Then $\lambda = F^{-1}(\Lambda)$, $\lambda' = F^{-1}(\Lambda')$ contain infinitely many z_n, resp. z_n', so for small r, C_r meets them at z_r, z_r' with $|F(z_r) - F(z_r')| > d$, contradicting Lemma 4.3. ∎

Lemma 4.5

The conformal map F extends to a continuous function from the closure of the disc to the closure of the polygon.

Proof. Define $F(z_0) = \lim_{z \to z_0} F(z)$ for $|z_0| = 1$ (Lemma 4.4). Given ϵ, there is δ with $|F(z) - F(z_0)| < \epsilon$ for $z \in \mathbb{D}$, $|z - z_0| < \delta$. If $|z| = 1$ and $|z - z_0| < \delta$, choose $w \in \mathbb{D}$ with $|F(z) - F(w)| < \epsilon$ and $|w - z_0| < \delta$; then $|F(z) - F(z_0)| \le |F(z) - F(w)| + |F(w) - F(z_0)| < 2\epsilon$. ∎

Proof of Theorem 4.2. So F extends continuously to $\overline{\mathbb{D}} \to \overline{P}$. The same argument applies to $G = F^{-1}$, since the property used ($C \cap \mathbb{D}$ is an arc for small circles C about boundary points) holds at every boundary point of P. If $z_k \to z \in \overline{\mathbb{D}}$, $G(F(z_k)) = z_k$ gives $G(F(z)) = z$; similarly $F(G(w)) = w$. ∎

4.4 The mapping formula

Let P be bounded by p with consecutive vertices $a_1, \dots, a_n$, $n \ge 3$, interior angles $\alpha_k\pi$, exterior angles $\beta_k\pi$, $\alpha_k + \beta_k = 1$; then $\sum \beta_k = 2$. Via $w = (i - z)/(i + z)$ $(\infty \mapsto -1)$, a conformal map $F : \mathbb{H} \to P$ (Riemann) extends to a continuous bijection of $\mathbb{R} \cup \{\infty\}$ onto p. First assume no

vertex corresponds to ∞: $F(A_k) = a_k$; after relabeling $A_1 < \cdots < A_n$, and F maps $[A_k, A_{k+1}]$ onto $[a_k, a_{k+1}]$ and $(-\infty, A_1] \cup [A_n, \infty)$ onto $[a_n, a_1]$.

Theorem 4.6 (Schwarz–Christoffel)

There are $c_1, c_2 \in \mathbb{C}$ with $F(z) = c_1 S(z) + c_2$, S the Schwarz–Christoffel integral of Section 4.2.

Proof. Let $1 < k < n$. F maps $[A_{k-1}, A_k]$ and $[A_k, A_{k+1}]$ onto two segments meeting at $a_k = F(A_k)$ at angle $\alpha_k \pi$. Choosing a branch of the logarithm, define

$$h_k(z) = (F(z) - a_k)^{1/\alpha_k}$$

on the half-strip of $\mathbb{H}$ over (A_{k-1}, A_{k+1}). h_k is continuous up to (A_{k-1}, A_{k+1}), mapping it into a straight segment L_k with $A_k \mapsto 0$, so by reflection it continues to the strip $A_{k-1} < \operatorname{Re}(z) < A_{k+1}$. $h_k' \neq 0$ there: in the upper half-strip,

$$\frac{F'(z)}{F(z) - F(A_k)} = \frac{1}{\alpha_k} \frac{h_k'(z)}{h_k(z)},$$

and $F'(z) \neq 0$ (Proposition 1.1), so $h_k' \neq 0$, and by reflection in the lower half-strip too. For $A_{k-1} < x < A_{k+1}$, h_k maps a small half-disc about x to one side of L_k injectively, hence (reflection) is injective on the whole disc, so $h_k'(x) \neq 0$.

Since $F = a_k + h_k^{\alpha_k}$, $F' = \alpha_k h_k^{\alpha_k - 1} h_k'$ and $F'' = \alpha_k(\alpha_k - 1) h_k^{\alpha_k - 2}(h_k')^2 + \alpha_k h_k^{\alpha_k - 1} h_k''$; as $h_k' \neq 0$ and h_k has a simple zero at A_k,

$$\frac{F''(z)}{F'(z)} = \frac{-\beta_k}{z - A_k} + E_k(z),$$

E_k holomorphic in the strip $A_{k-1} < \operatorname{Re}(z) < A_{k+1}$. Similarly $\frac{F''}{F'} = \frac{-\beta_1}{z - A_1} + E_1$ for $\operatorname{Re}(z) < A_2$ and $\frac{F''}{F'} = \frac{-\beta_n}{z - A_n} + E_n$ for $\operatorname{Re}(z) > A_{n-1}$. F also reflects across $(-\infty, A_1) \cup (A_n, \infty)$, so it continues boundedly to $|z| > R$, hence is holomorphic at infinity: $F(z) = c_0 + \frac{c_1}{z} + \cdots$, and $F''/F' = O(1/z)$. As the strips cover $\mathbb{C}$,

$$\frac{F''(z)}{F'(z)} + \sum_{k=1}^{n} \frac{\beta_k}{z - A_k}$$

is entire and vanishes at infinity; by Liouville it is zero. With $Q(z) = (z - A_1)^{-\beta_1} \cdots (z - A_n)^{-\beta_n}$, $\frac{Q'}{Q} = -\sum_k \frac{\beta_k}{z - A_k} = \frac{F''}{F'}$, so $\frac{d}{dz}\left(\frac{F'(z)}{Q(z)}\right) = 0$ and $F' = cQ$. A final integration yields the theorem. ∎

Theorem 4.7

If a conformal $F : \mathbb{H} \to P$ maps $A_1, \ldots, A_{n-1}, \infty$ to the vertices of p, there are constants C_1, C_2 with

$$F(z) = C_1 \int_0^z \frac{d\zeta}{(\zeta - A_1)^{\beta_1} \cdots (\zeta - A_{n-1})^{\beta_{n-1}}} + C_2.$$

So the formula is obtained by deleting the last factor of the Schwarz–Christoffel integral.

Proof. Translate so that $A_j \neq 0$. Choose a real $A_n^* > 0$ (different from the others; [Supplement]), $\Phi(z) = A_n^* - \frac{1}{z} \in \operatorname{Aut}(\mathbb{H})$, $A_k^* = \Phi(A_k)$, $A_n^* = \Phi(\infty)$. Then $(F \circ \Phi^{-1})(A_k^*) = a_k$, $k = 1, \ldots, n$, and by Theorem 4.6

$$(F \circ \Phi^{-1})(z') = C_1' \int_0^{z'} \frac{d\zeta}{(\zeta - A_1^*)^{\beta_1} \cdots (\zeta - A_n^*)^{\beta_n}} + C_2'.$$

Change variables $\zeta = \Phi(w)$, $d\zeta = dw/w^2$; since $\zeta - A_n^* = -1/w$ and $\zeta - A_k^* = (w(A_n^* - A_k^*) - 1)/w$, and $2 = \beta_1 + \cdots + \beta_n$, the integrand becomes $C \prod_{k<n} (w - 1/(A_n^* - A_k^*))^{-\beta_k}$ up to a constant factor. Since $1/(A_n^* - A_k^*) = A_k$, setting $\Phi^{-1}(z') = z$ gives the formula. ∎

4.5 Return to elliptic integrals

$I(z) = \int_0^z \frac{d\zeta}{[(1-\zeta^2)(1-k^2\zeta^2)]^{1/2}}$, $0 < k < 1$, maps $\mathbb{R}$ onto the rectangle R with vertices $-K, K, K + iK', -K + iK'$ (Example 3); we show it maps $\mathbb{H}$ conformally onto R. By Theorem 4.6 there is a conformal $F : \mathbb{H} \to R$ sending four real points to the vertices, and after an automorphism $-1, 0, 1 \mapsto -K, 0, K$. Let $F(1/\ell) = K + iK'$, $0 < \ell < 1$. $\tilde{F}(z) = -\overline{F(-\overline{z})}$ is also such a map (symmetry of R), so $F^{-1} \circ \tilde{F}$ fixes $-1, 0, 1$, is the identity, and $F(-1/\ell) = -K + iK'$. By Theorem 4.6,

$$F(z) = c_1 \int_0^z \frac{d\zeta}{[(1-\zeta^2)(1-\ell^2\zeta^2)]^{1/2}} + c_2.$$

$z = 0$ gives $c_2 = 0$; $z = 1$, $1/\ell$ give $K(k) = c_1 K(\ell)$, $K'(k) = c_1 K'(\ell)$, with

$$K(k) = \int_0^1 \frac{dx}{[(1-x^2)(1-k^2x^2)]^{1/2}}, \qquad K'(k) = \int_1^{1/k} \frac{dx}{[(x^2-1)(1-k^2x^2)]^{1/2}}.$$

K is strictly increasing and $K'(k) = K(\tilde{k})$, $\tilde{k}^2 = 1 - k^2$, is strictly decreasing, so K/K' is strictly increasing; $K(k)/K'(k) = K(\ell)/K'(\ell)$ forces $k = \ell$, $c_1 = 1$, $I = F$.

Let sn be the inverse of I (Jacobi's notation, cf. sin); it maps $R_0 = R$ onto the closed upper half-plane. Reflecting across the lower sides gives rectangles $R_1, R_2, \ldots$ with $\mathrm{sn}(z) = \overline{\mathrm{sn}(\overline{z})}$ on R_1, $\mathrm{sn}(z) = \mathrm{sn}(-iK' + \overline{z})$ on R_2, and so on, extending sn to $-K < \mathrm{Re}(z) < K$ with period $2iK'$. Reflecting in horizontal rows extends it to $\mathbb{C}$ with period $4K$. So sn is *doubly periodic* (periods $4K$, $2iK'$; meromorphic, two simple poles per period parallelogram): Chapter 9.